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Basic Static Timing Analysis

Course Version 3.0

Lab Manual **Revision 3.0**

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Module 9: Analyzing a Timing Report

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Lab 9-1 Setting Up the Database

Objective: To download and set up the course database.

Software Requirements

You can use any synthesis tool to load the constraints, run the synthesis, and analyze the results.

This lab was built for the following software:

- Tempus™ Timing Signoff Solution software

Linux Instructions for the Database

The software for this course only runs on Linux, so there are no PC or UNIX instructions for the database.

After downloading the database file, uncompress it using the following command:

- gunzip BSTA_*.tar.gz
- tar xvf BSTA_*.tar

These commands create a directory called *bsta_labs*.

Make sure that you read the *README* file in the *bsta_labs* directory concerning any specific sub-versions of the software to use with these labs.



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Lab 9-2 Creating Constraints

Objective: To write your constraints for a design in a manual format.

In this lab, you edit a *constraints.sdc* file and add constraints according to a design specification. After editing the file, compare it with the provided solution.

Setting Constraints

1. Change to the *work* directory:

   ```
   cd bsta_labs/work
   ```

2. Use the following specification to write your constraints into the *constraints.sdc* file.

   Note: Use your favorite text editor to edit the *constraints.sdc* file.

Operating Condition	**slow**
Wireload Mode	top
Wireload Model	custom_high
Port	Specification
xxin_ORSCLK	▪ Clock Name: ORSCLK ▪ Period: 5 ns ▪ Waveform: {0 2.5} (50% duty cycle)
FtiCoreClk	▪ Clock Name: CORECLK ▪ Period: 5 ns ▪ Waveform: {0 2.5} (50% duty cycle)
All inputs	▪ Driving cell: INVX1 ▪ Input delay: 1 ns ▪ Related clock: CORECLK
Reset, xxin_ORSCLK, and FtiCoreClk,	▪ Input delay: 0 ns
All outputs	▪ Output delay: 1 ns ▪ Capacitive load of 300 femtofarads (not picofarads) ▪ Related clock: CORECLK
All clocks	▪ Skew: 100 ps ▪ Clock jitter: 0 ns

3. Save the constraints.*sdc* file.

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Solution

Compare the constraints you have created with the following:

```
set_operating_conditions -library slow
set_wire_load_model -name "custom_high"
set_wire_load_mode "top"
set ALL_IN_PORTS [all_inputs]
set OUT_PORTS [all_outputs]
set_driving_cell -library slow -lib_cell INVX1 $ALL_IN_PORTS
set_load 0.3 $OUT_PORTS
create_clock -name ORSCLK -period 5 -waveform { 0 2.5} [get_ports
   xxin_ORSCLK]
create_clock -name CORECLK -period 5 -waveform { 0 2.5} [get_ports
   FtiCoreClk]
set_input_delay 1 -clock CORECLK $ALL_IN_PORTS
set_input_delay 0 -clock CORECLK [get_ports *Reset*]
set_output_delay 1 -clock CORECLK $OUT_PORTS
set_clock_uncertainty 0.1  [all_clocks]
```

Note: These constraints would not match the constraints in the *solutions* folder. They were used to create the netlist for use in the next lab.

Testing Your Constraints for Synthesis (Optional)

To test the constraints, you can run Cadence® Genus™ Synthesis Solution.

1. Start Genus.

   ```
   genus
   ```

2. At the Genus prompt, run the commands:

   ```
   read_libs -max_libs  {../libs/liberty/slow.lib
      ../libs/liberty/wireload.lib}
   include filelist.tcl
   read_hdl $vfile
   elaborate top
   read_sdc constraints.sdc
      //OR
      //read_sdc solutions/constraints.sdc
   ```

   Are the constraints clean?

   Answer: _____

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Summary

This exercise gives you a good idea of what you need to do to write your constraints.



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Lab 9-3 Analyzing the Timing in a Design

Objective: To analyze timing using the timing reports.

In this lab, you use the information in a timing report to debug a design that is not meeting timing.

Generating Timing Reports

In this section, you run Tempus software to generate the necessary timing reports for analysis.

1. Change to the *work* directory:

   ```
   cd bsta_labs/work
   ```

2. Run the following command to start the Tempus Timing Signoff Solution and load the necessary reports for timing:

   ```
   tempus -stylus –files run_tempus.tcl
   ```

   The *run_tempus.tcl* file has commands to run the software. Look at the file in a separate window.

   Make sure that Tempus software is installed and is in your path, and the software is licensed.

Analyzing a Timing Report

1. Generate a timing report to see if your design meets timing by entering:

   ```
   report_timing
   ```

   By default, only one worst/most critical path is reported.

   Note: To report additional paths, you can use the *-max_paths <integer>* option.

   You can redirect this report into a file using “>” and look at the file instead.

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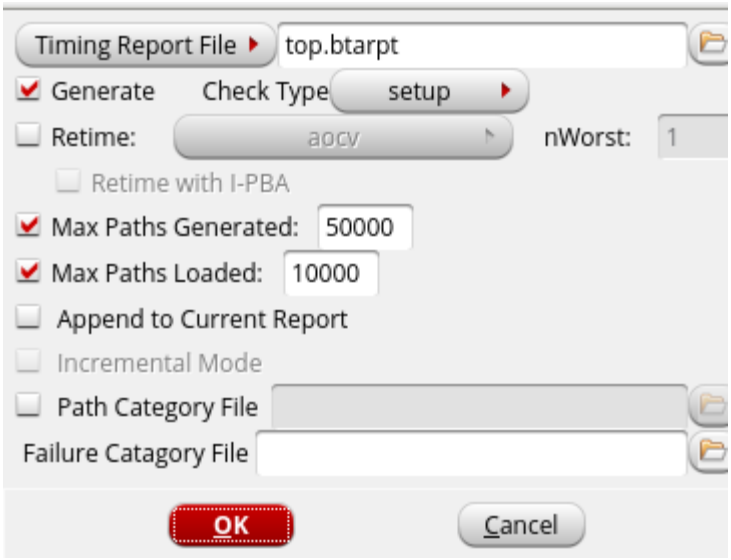
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Alternatively, you can generate a **Setup** timing report in the graphical interface as follows.

a. Open the graphical interface by entering the *gui_show* command.

b. In the graphical interface, click on the plus icon and open the **Analysis** tab.

c. Click the Browse icon.



d. With the default selections, click **OK**.

2. Analyze the timing report.

   Did it meet *timing? If not, does anything stand out as a problem?*

   Answer: Yes/No_____, _____

3. Continue analyzing the report by answering these questions:

   What is the slack of this reported worst/most critical path?

   Answer: _____ ps

   What is the start point of this critical path?

   Answer: _____

   What is the end point of this critical path?

   Answer: _____

   What type of path is this? (Reg-Reg, Input-Reg, Reg-Out)

   Answer: _____

   What clocks drive the start point and the end point?

   Answer: _____

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Locate where the insertion delay was reported. Did you find it?

Answer: Yes/No_____

Tip: Find it in the slack calculation section.

What are the cells with the largest delays?

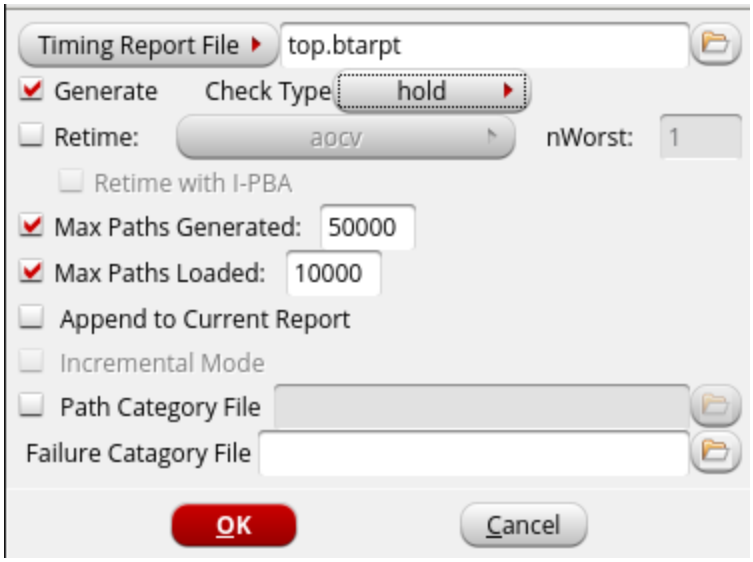
Answer: _____

Tip: Find it in the arrival time calculation table.

Verify Hold Timing

Let us find out if any other paths are violating.

1. Generate a **Hold** timing report.
   a. In the graphical interface, click on the plus and open the **Analysis** tab.
   b. Click Browse icon.
   c. Change the check type to **hold**.



   d. Click **OK**.

How many other paths violate timing?

Answer: _____

Do all these paths have the same endpoint?

Answer: Yes/No_____

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Verifying Multicycle Paths

You should see that there are multicycle path constraints reported in the report.

1. View the timing report.

   This report shows multicycle paths that are from the *GlCOreSoftResetSynR00C* port.

   What is the slack reported?

   Answer: _____

2. Edit the *DESIGN/top._default_constraint_mode_.sdc* file.

3. Look for *set_multicycle_path* commands in the file.

4. Remove the comments on the hold multicycle path from the file.

5. **Save** the file.

6. Reset the timing and regenerate timing reports by entering:

   ```
   read_sdc DESIGN/top._default_constraint_mode_.sdc
   update_timing -full
   ```

   This command will reset the SDC that was previously read in and then read the SDC file again (the one you updated and saved).

   Did the timing of the path get any better?

   Answer: _____

Additional Steps (Optional)

1. Look at the following timing report:

   ```
   report_timing -to U_fticustomc
   ```

   Is there time borrowing in this report?

   Answer: _____

   Latches can borrow *time from other cycles* to meet the timing of the path.

2. Feel free to modify the constraints, see the effects of the changes in the tool, and then run reports.

3. Exit the software by entering:

   ```
   exit
   ```

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Summary

In this lab, we looked at how to look at different timing reports in Tempus software and the effects of the constraints on timing closure.



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